<<S.M.H.Mousavi_Patent_Summary>>

Patent Title: **Portable system for monitoring emotional state and concentration in interviews based on multi-modal fusion of visual and audio data on a single-board computer platform**.

This patent relates to an intelligent multi-channel system for monitoring and analyzing an individual's emotional state, stress level, and focus based on the simultaneous processing of visual and audio data. The primary limitation of existing systems is their reliance on only one or two data sources, leading to an inability to provide stable and reliable analysis in real-world conditions.

In this invention, data is extracted from multiple channels, including facial imagery, body movement, mouth movement, gaze direction and eye movement, blinking, audio signals, and chest movement. Following temporal synchronization, the data is processed frame-by-frame to extract channel-specific features. Subsequently, a multi-channel fusion mechanism with adaptive weighting generates the final composite signal.

The system is capable of calculating emotional and cognitive indices in real-time, storing the data in structured files, and visually displaying the results directly on the live video feed. This invention provides an integrated, contactless technical solution for analyzing psychological and behavioral states.